

Active Inference Insights 024 ~ Darius Parvizi-Wayne (Season Finale // Reflection)

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all right it is July 8th 2024 and it is active inference insights number 24 a reverse Insight so welcome Darius how are you doing very well I apologize everyone say for my stupid headband I'm currently in El Salvador um and it's very warm and my hair is everywhere so I've just I've decided to strap it in so um it's a bit of it's a bit of a look I'm very well how are you Daniel good well as we just briefly discussed this is sort of the season finale slash pivot point or next jump and we will discuss some of the Inception and journey of insight series The 1 through 23 main episodes what is next for insights what is next for you and just however long that is so let us begin with Y insights how did it come to be and what were the beginning steps like well first it's annoying that we never made this to 25 you know it would be nice to have a nice square number but 24 will do that's life I guess um how's it come back well firstly thanks to your generosity um in many ways and the institute's generosity so I'm very grateful for that um i' had done podcasting with of University a little bit after but I think I just noticed there was there was a sort of Gap at least in the institute's offerings uh the live streams not like this one but the other ones can be quite Technical and I don't think necessarily offer a way in and a way to see how active inference exists in the sort of context of theories of biology or Neuroscience or philosophy or whatnot so there was a space there I thought and I just I guess I guess it was also this idea that it was a way for people to learn alongside me um and I think that's always quite an efficient and effective way of podcasting uh because I don't you know there's one thing that where people know nothing and it's very difficult to have a conversation with an expert and there's another where there two experts speaking and it's inaccessible for the audience so was um I think we hopefully found a kind of sweet spot where I was sufficiently knowledgeable albeit to a quite limited degree at the beginning I'd say uh but enough that I could ask insightful questions and be curious the way it came about really was I did my masters at UCL in 202 from 2022 to 2023 and about halfway through I sort of discovered active inference probably through predictive processing because obviously Carl is

at UCL and it's a big Hub fact of inference although I think there's an interesting relationship between active inference in UCL because not everyone's necessarily on board and then I started attending I got on the Discord so Miles who we just mentioned before Miles Taft was my supervisor at UCL was also and is also interested in active inference and he recommended the Discord and the The Institute so I started doing that started attending the textbook groups and then I think I just messaged you um when would that be September October and it all came together quite nicely and rather quickly and then then again just people to thank Carl really first and foremost to be the first guest um a good three hours with Carl was like very nerve-wracking experience but I think a pretty you know throw yourself in the deep end and then I think hopefully it's just gone from strength to strength um and I'm sure we can sort of talk about how it's evolved one thing about that certainly by view count which is not everything the most popular videos on our Channel over the last year have been basically Chris Fields's 2023 physics' information processing and your series and now I'm kind of reflecting on that and uh when we have a presentation like a guest stream that can get technical or a live stream with a paper Focus there are people there it's human however it's also like a conversation with literature and literature in conversation with other topics whereas in Chris Fields's case with the course like format that's a uh similar to a professor to audience and then what you kind of brought into play was like also the person to person and so that sort of like direct human element of the conversation which is much more like a textbook group but of course a textbook group still has the textbook group as reference point but once you kind of um swim away from the wall of any text then there's this like wide space of what you can ask people and and like how you prepared for sure and I think people naturally are attracted to the kind of uh you know what John Bal di Loos um this kind of mutual dialectical unfolding where I don't know we're both we're both um affording each other the opportunities for insights and one thing I always sort of wanted to do and never got around to doing was doing in-person insights episodes um there was the possibility of doing that I mean I think the more it can be in the more it can be enacted the more it can be situated the better and maybe that's the kind of idea going forward I mean I recognize also that one of the benefits of The Institute is it's kind of its location is everywhere and nowhere it's this distributed map and that's really amazing because I think what you see on the Institute is all these people coming from completely different parts of the world with completely different interests and um being unified over this one over an interest in this framework at the same time I think yeah hopefully what what what what sort of developed was this kind of mutual unfolding and I think that can

really only be done in dialogue I think you have to be quite an exceptional speaker an orator to be able to do that just through monologue yeah wow interesting point there okay so Carl and then dot dot dot dot dot so yeah do any come to mind like what was sort of a um a leadup or a followup that was surprising or you know without naming innocent how how did people differ and how they approach like what you ask and did well look everyone to be honest everyone was really amazing I mean for me just like being a pre-doc but being like to have the that generosity of people to give me their time and actually re and take it seriously and not give uh you know flippant or or oversimplified answers was really something and I think I recognized that with call when I asked him about active influence he gave me a sort 15 minute answer which basically maths and I thought okay well this is not going to be this is not going to be a soft launch um so everyone sort of unified has taken it really seriously and been really engaged and that was amazing I think there were a couple Pinch Me moments um so one like I've always loved John fi um really since discovering Awakening from the meaning crisis probably when I was about 17 or 18 he's always been a massive figure in my life and I was lucky to meet him last year and to be able to speak to him was a really bizarre and special moment so yeah John was really amazing I think also there were some people who I think what was nice about the process was integrating some people who weren't necessarily active influence per se but still had insights from different disciplines and different domains which could bear on act influence so I really enjoy speaking to Mar and stendera um about haiger because this kind of notion of us being being directed ultimately towards a Futurity I think is very ining with active inference in this emphasis on expected free energy but everyone's been yeah everyone's been a total Delight I think also what's been really fun is having like the debate was really fun um about the markco blanket trick and I think that's another thing to say I really enjoyed having people who weren't necessarily not only people from different domains but also people who weren't necessarily in the camp of active inference so whether that was you know a crystal leer for example just being able to probe the relevant questions I think is really important because the the the framework is always evolving and it's always important in my opinion at least to have these not necessarily outside voices but concerned voices just to temper uh proceedings and I guess you had that when you know you've done live streams with people like Mel Andrews um so I think you know if this something like this continues it's important that we break out and listen to other voices who might have something to say awesome thank you yeah well about the sort of contents what were some topical themes whether ones that developed like even over those months or just some emergent themes or like

what topics did you find coming up or developing yeah so I think so I'm not by training a philosopher of Science and so I guess when I discovered that to inference this whole notion of the map and the territory was not Forefront in my mind um and in many ways my enthusiasm fact of inference I think has been quite efficiently and importantly mitigated or mitigated but just constrained by that conversation so to what degree can the maths actually really tell you tell us anything about lived well one lived experience two human behavior and three like what the brain is actually doing or the brain or the body what they're actually doing so that was a recurrent conversation that kept coming up over and over again uh this map territory or map territory um and I think hopefully it's been a contributor contributor to this wider conversation that I know is happening within active inference and The Wider Community again about what active inference can really say I tended a talk by to Schick last week about well I think it was predictive processing flirt with transcendental idealism but the conversation turned into a kind of map territory thing and I think a lot you know now and even people coming into active influences a new field are very much aware of the importance of that conversation so that was one thing that I think was really really helpful the other thing that always kept coming out was Consciousness um and the lived experience and I think that's in part because that's really what my um research focuses on and so people may have noticed that the podcast was slowly transforming into a podcast on Consciousness but so be it um and in that case I think all that was cemented was the fact that I don't think we know very much at all um so I mean that in terms of the metaphysical question really the hard problem and then I guess on the flip side of that is the importance of taking lived experience very seriously and in my opinion always kind of starting with the phenomena doing the intricate and careful phenomenological analysis of the of the actual phenomenon in question before applying your models or running your experiments and so on so yeah the map territory Consciousness um and and and much more and much much more and I think also just active influences plays in the history of ideas um whether you are coming from a kind of computational stance you're thinking about hel HS machines or Jeffrey Hinton or its relationship to neuron Nets or whatever or whether you're a phenomenologist or whether you're an analytic philosopher of mind like Thomas metzinger just recognizing that the history and the future is has come and will come after us and it's I guess always important to think about our place within that that that stream of time nice but there was much more there was much more oh yes I'll definitely look forward to not just Rel listening but also to jointly analyzing the core entities and the active inference ontology okay two questions one and two take as long as you want one what is active inference two why is it

important to have insights about active inference so I think my wife I might have just put it there but what's the first question what is AC of inference and the second one why does it matter to have insights yeah so active inference is as we all say the corollary of the free energy principle it's um it's not the same as predictive processing and I think that's an important um proviso although a lot of people conflate the two so the free energy principle um tells you in terms of mathematics uh that an organism or an entity that persists over time can always be described as if it is minimizing this upper bound on surprise that we call variational free energy um and then act of inference I've always I I think to be fair there's an interesting meta question here about exactly what we are talking about when we're talking about active inference but I've always thought of that as the process Theory associated with free energy minimization now this as if or can be described as the is the kind of Paramount thing here um because yes indeed there's a mathematical way that you could start from like a [___] plank equation or or or any sort of um start from your Theory of physics which is that it's a um it's a whatever it is you know you have your mathematical equation and then you you say okay well therefore given this we can cast self organization as this minimization of variational free energy the question is okay well we make the axiomatic assumption that things do persist over time so that's that's relevant um but then the question really is okay given that things are persisting over time and given that they can like that is a really nice Launchpad how are things actually doing that and then I guess this is the overlap with things like autoparesis and in activism and cybernetics and all of this stuff why is it relevant to study AC well I think because we we have to take seriously the notion that things are self organizing um it's not a trivial fact right this is I think Schrodinger's big Insight um back in the 50s which was that life is a quite miraculous phenomenon it is this neg entropic phenomenon that seems to defy the second law of Thermodynamics um or at least offer a quite interesting coun example to this and there I know again this is not necessarily my field of expertise but I know there's a broad domain of thinking why that happens in the first place I mean why are these pockets of neg entr entropy in the universe anyway um and this obviously can take you all the way to stuff like arguments from design um all the way back down to like the role of lived experience so I think it's very important to study self-organization and but that said it's important to realize that the mathematics won't take you all the way there so for me I've always viewed AC of inference in the frage principle as a very interesting Launchpad but always with the recognition that the answer I don't think is is it can't be in the mathematics is the mathematics at least the way that I've of always viewed it or or come to view it is an instrumental Tool uh a

very insightful instrumental tool but not as we say the territory itself and I know Maxwell who I just chatted with a couple of days ago he's just released this preprint with dtton and Carl and they make very explicitly that point so I think the next steps in active inference is going to be something like so what is so what is the point right like you have the math and it's this really interesting described tool but then what and is it is it really how secluded is it from what the brain is doing because I think we forget that within the whole Act of inference ecosystem although it's different you do get these ideas coming in like predictive processing or predictive coding or discrete state space PDP work and that's much more of a process theory about it actually what is the brain doing and then you know within those theories you start having the integration of stuff like active inference oh oh sorry variational free energy and expected free energy so for me active inference is really interesting because one thing seems to be true which is that we are self organizing and self-maintaining and self-distinguishing I think the real work needs to be done on saying what is the instrumental tool what is actually happening in a realist Manner and where are the where are the where how do we get from one to the other and obviously this has been the Paramount question throughout but I think it's high time that someone comes in and really asks the the the sort of the path it's like a generative passage but for the for the science real active inference has never been tried well well well yeah well it was a stochastic differential equation which was which was escaping my uh attention um what that very interesting account um impresses on me is you started with the downstream relationship of active inference to free energy principles so that's the high road yeah and in that sense there's a grounding in general thingness particular partition more like a systems theory then you said that's a Launchpad to get down to the meeting point because you get that surprisal minimization VF from the top and then with all of these existing basian architectures as various low road paths those also have kind of convergently described the evidence lower bounds multiple V using it's actually coming from the low road methodologically but it's as if it's coming from the high road so that's one place they meet then you said well that is like that's like another orthogonal Launchpad for people who want to do something other than develop that infrastructure which will remain its own research program and Direction and everything but then it's going to be that's so what so what okay so we have RX INF fur with the dragon drop and everything works fine and here and then what so that's that's very interesting and anticipatory I think I think just think first and foremost always for anyone coming in just recognizing that these things don't all mean the same thing the free energy principle does not mean predictive processing predictive processing does not mean a you

know it doesn't necessarily rely on an upper bound on surprisal and so on and so forth and so I think I think all credit to P Petula and Friston for sketching this out quite effectively in their textbook and also the way that you guys teach this in in the textbook group and then also the the resources of The Institute I don't you know it's important I think I mean there are lots of Twitter debates that rage about how ignorant the active influence Community is interestingly within the kind of ecosystem they're not uh and they're very I I personally think they're very humble and aware of the of the limitations so yeah so so I don't know what the ultimate solution and the symphasis is whether there is such a thing um but I think it's it's it's a kind of we got to start with the multiplicity before looking at the unity and then trying to see if there's un unity in the multiplicity which is sort of both Mill Christ and he and all together you know and alluded to in the neat and Scruffy also early in the textbook um okay so that that that's very interesting um so several paths ahead yes first how about some of your own directions and then we'll talk more about like where it could insights go sure so I'm starting my PhD in October in at mcarl in Sydney so I'm my supervisor in Nez poito and Raphael Mia and I'm very very excited and very grateful to have the opportunity uh and moving to the other side of the world as well which is I will be I am on the other side of the world right now but you know actually setting up residence it's slightly scary but it's exciting nevertheless um and I want to explore principally subjectivity and the sense of self as well as what if like what a self is um so this some of this is foreshadowed in the work that we did on Flow State so this was uh me L KL Ry jaob and Miles TFT and I think there what became really Paramount to me was that the notion of self almost like the notion of active inference becomes conflated and confused because people don't make it clear where they're talking about the metaphysical self as if there is a sort of cartian ego or a sense of self even when they're talking about sense of self they're not distinguishing between a a self as subject and a self or a sense of self as subject and the sense of self as object so I I'm really interested in subjectivity the fact that experience seems to have a certain for meanus and there's an ongoing debate about the degree to which that's a universal condition of Consciousness so this is people like Anica but really going all the way back to Dan Harvey and um Joseph haras and Louis SAS all the way you know back into the '90s and I think linking all that together as well is an interest in being being in the world in the hiigaran sense and skillful coping in the sort of Dr fussy and Merlo Ponte sense so seeing if there's interesting connections between subjectivity and this familiar coping that we have in the world so that's a couple of strands and then I think naturally from that one becomes embroiled in what is being in the world to what degree does it um involve representations

there was just this big paper that came out led by Johannes Jer on the fact that relevance realization and agency and cognition therefore can't be computational so for me I read that and I think it's very Merlo ponan and very dra foran um and so those are the big philosophers that kind of inform my thinking on this front so just seeing I mean it's all quite open but I think hopefully there's a kind of cluster there that's emerging because there is also a more meta question about the degree which we could computationally model something like subjectivity if the fundamental aspects of cognition aren't computational algorithmic and representational and I think it's a fair presumption that subjectivity might not be so this is a big mystery basically that no one knows where any of where the four meus comes from and first and foremost no one really necessarily knows what the four meus is so some people take almost without critical analysis that the for meanus is because there is self a homunculus located behind the Eyes For Whom experience is given I think it's quite clear that that isn't the case from the human analysis from the neuroscientific analysis so there's really nice work by people like Kenneth willford um and going all the way back to Aaron goit and SRA about the fact that Consciousness might be the the subject relas of Consciousness might be Consciousness itself so it's exploring this kind of thing and then I then I guess for me as well something that's really important is collaboration um I've been really really really really lucky to be coll to have collaborated already with really amazing people um not only the people on the flow paper but also people like Libby s on the piece we did on tourettes and self modeling so really just trying to get integrated into this community this philosophical this um theoretical biological community as much as possible and then seeing again what act of influence can do for all of that so I think not necessarily making active inference the front and center because I not sure I don't think I could without necessarily because of all of the big meta scientific questions that reside in active inference the map territory thing so I think using it as a tool and then if there are developments either done by myself or other people about a more biologically realist robust theory of actually what the brain is doing but absolutely fleshing that out and trying to do this what Maxwell and of called the generative passage tying phenomenology or or the lived experience to modeling to experimentation but always always being aware of just how fraught with danger or how precarious and fragile that passage can be so that's the idea at least it's quite broad it will narrow down cool wow well that's it's exciting to begin the PhD um what you said about active inference not being front and center for some reason it made me think of like famous or powerful photography or documentation and it's like the map and the methodology so that' be like the open source methods element however outside of the gear

enthusiasts connoisseur respector development Community it'd be like okay but what about the picture so so not that that's a reductionist view on the totality of like photography or anything like that but just like it it may become relevant for for a larger and larger fraction to focus on how that picture looks rather than um that's absolutely right what the methods and the tools are I think just I just think maybe because of the way I'm composed and maybe because my background is ultimately in some in philosophy of mind and philosophy I I I do think for myself starting with the phenomena is the first place never for not not only never forgetting the phenomena but starting with a phenomena so I've just been writing this thing about attention and it's funny how strange the notion of attention can be once you lose sight of how it actually phenomenologically unfolds and and how it structures Consciousness so for me and again this is not disrespecting anyone who wants to start with the modeling or wants to start with the map but I'm probably inclined to go the other way that makes me think about textbook group it's like quite literally sometimes circle square Line Circle Square o s a b c d e and then it's like well even if you were going to make a map of your street and intersection you'd be talking about like 20 to 100 circles and squares pulled up to the level of abstraction of something like a PDP and then that's like getting into the multi-agent and all of these Concepts yeah and then also made me think of like what is this s obviously great question to um to do the PHD in and the immune system and the sensory and the effector and all these different self Concepts and then seeing what's happening now with like agentic ai ai agents defined in terms of like perception and action which is like one thingness and then at the same time as as Maxwell also emphasized in the uh Discord recently like the agentic focus which is one system of Interest or genre of system of Interest actually minimizes the apparent scope because especially grounded more in the high road of just how things are as if they are rather than any particularly built architecture then it's like yeah there's no reason why it should be kind of an agentic boundary it could be like an ecosystem of shared intelligence and then that's where all these self concepts are in like a kind of Multiplicity almost by definition sure I mean I I I guess I guess ultimately we don't know um so so like this notion of individuated s is is in many ways at least I've always taken it to be an underlying axiom of active inference right I mean if one was to say that well we're all just non- individuated things in the soup then you can't really apply the Mark or blanket partition in the first place so it's a healthy way to start but even if you read I mean I was just reading something yesterday on it was like autopoietic autonomic and activism um so just just it it was a really interesting piece see if I can find who wrote it it's a very it was very good um but ultimately and they were they were trying to do is

saying okay why is in activism why why are there been these two strands for in activism sensory motor in activism and then autonomic based on basically autopoietic in activism and trying to kind of merge the two and saying okay well what's at the heart of living might be something like sensory motor and activism but not discounting the fact that there is this operational closure within the system but again one has to recognize there that like there's a there's a kind of embedded circularity which is that we already take the system for granted so we're saying like what what can what allows the system to self-maintain whether it's operational closure but thermodynamic openness and then but what does that do well it maintains the self-organization and and allows for self-maintenance and so there is and and this is obviously recognized by those authors there is this kind of inherent circularity and this notion of like a strange Loop has clearly always been pointed to at the heart of of of well intelligence and cognition so I think also getting to the to the bottom of that Circle which for many philosophers would be problematic but actually for these theorists is kind of at the very heart of their theories is a really interesting thing but yeah again not forgetting that literally recognizing or believing theoretically that there are individuated things can look like an axiomatic presupposition and in active inference it almost certainly is one so yeah I mean I mean what is a self really is the big question I guess there's there's really good Pap I always do this it's like I'm in podcast mode again but it's um verel I think it's 1991 it's called a meshwork of selfless selves really good really really good because again it's not saying that this is a self in the sort of strict cartesian way that we all might have a focus psychology theory of but like some of these are intuitive or viable ways that we could Define the self whether this is a sensory motor boundary a Cellular boundary and so on and so forth so yeah it's it's a surprisingly elusive concept the self yeah so much to say on that I mean even if you apply a utilitarian or pragmatist approach to your own agency over an intentional profile over self-concepts it would be like that's the strangeness or part of the strangeness and the second order with the okay I'll read some comments from the live chat and we'll keep talking about this before we do eventually talk about like where the series go okay upcycle Club writes greetings wave hand emoji I have really enjoyed the insights active inference series with Darius I wish you all the best in your future endeavors well thank you that's very kind I mean yeah thank yeah go go go oh yeah thanks Up Cycle all right Terry mold writes I've loved the talks the latest one I really thought the free energy principle could resolve the different approaches if you remove the idea of a pre-ordained agency of the individual make sense um yeah I mean again depends on how you define agency but um if you just Define agency as the ability to again in like the in activist

tradition to set your own norms and standards with respect to your surrounding mure um then active INF is very consonant with those ideas so I'm not too sure whether you can really dis uh you know discard the notion of agency when you're talking about active inference because at the very heart of active inference is this sensory motor contingency so again I think it slightly depends on what one means by agency um if I mean clearly there are things that don't necessarily move and change the world according to their internal norms and by this I don't mean conscious norms I mean the Norms that an eoli bacterion would have and for those I guess at the you know at the mark of blanket you just leave the the active partition no but no I mean I mean I don't think active influence completely flens out this distinction between agentic and non- agentic systems um in many ways it might reinforce this idea that agency is active influence and active influence is agency so again always just a epistemic humility I guess I don't know what AC I don't know what the free energy principle solves but I don't think it it's going to be able to uh yeah discard and dispense of agency so so sort of Thoroughly okay Terry added like an emotional valtion which I think gets added after the event well what what I think is interesting about this and relates to your flow work and some of the other discussions that have come up about time is that view from the inside and view from the outside of potentially time scales that are working in this integrated way where there's this kind of like it's it's a complex moment what is happening and that's even before getting into like different ways we can think about SpaceTime so it's like in that rewriting pre-writing now casting body integrating Zone there's so much to explore and and like you said about the map territory earlier it's like good to go into a complex territory knowing what's a map and what's the territory at least in kind 100% And so whenever you know we there is a there is a PM DP in the float States paper and in The Tourette's paper but hopefully it's made pretty clear that this is an idealization um I was very lucky actually as an undergrad to study Chomsky in linguistics and the notion of idealization comes up a lot in Chomsky and so I think I was at least in that sense quite well I know I said earlier I'm not a philosopher of science but there was some we had there there were quite a few discussions about idealization and science so yeah yeah in those cases I mean God knows how recursive and hierarchical these layers are and I think in some cases all you have to do is point at it and say you know at this degree of complexity one must remain silent there's an interesting thing there about um emotion coming after the um after the phenomenon I'm not quite sure I agree I mean I have flirted with the idea that for example in flow states there is no consciousness um or at least no emotion that there is an ongoing discussion again I'm not I'm not an emotion theorist but

there's a I believe it's within appraisal theory about the fact that just if we take a motion to be an appraisal of let's say an inter receptive signal it doesn't actually have to mean that this can only be done offline it could be done in real time and I think informed Again by someone like draus or Merlo Ponte that's that's quite right um the emotion in many ways whether it's now there's an interesting question here is about actually what the causal role of that emotion is because there are two different things here one is its causal role and one is its temporal um status the interesting uh causal role is whether that emotion actually does anything for the organism who's trying to return to this kind of equilibrium or what um draus would call your sort of optimal grip with the gal so I don't know I I don't think the flowing agent is non conscious I don't think they're a robot um so no UL to answer your question I don't think the free energy principle solves everything by dispensing with agency because one I don't think it can answer the question of agency two I don't think there's a robust definition of agency within the free energy principle literature or elsewhere and and thirdly under certain definitions agency is inbuilt within actively inferring agents I'm note that I I'm using agents with speech marks because when I talk about asan I'm not talking about some cartisian homunculus so big questions big questions um which need lots of years of deep delegation to to resolve cool okay more to the operational closure yes so having organized many many live streams and worked with many people what was very different about your co-organizing was basically you did all the invitation you recorded it and just shared the upload so logistically what did it take in terms of research background prep confirmation all that kind of stuff how would you evaluate that and or how how would you we'll just ask that question first just how was it how was it and how you actually did it well I I like doing this stuff so I love emails um and maybe this is a betrayal of my age and inexperience but I I love sending emails so well you very kindly had the spread we have a spreadsheet which is going to be hidden from everyone else but we have a of Coda spreadsheet where we have just a list I think there was there was a list of people who had been on live streams before and then maybe potential guests and then we we did another one where it was people and then there was a way of sort of categorizing it people I'd invited people I was thinking invited people had been on so for me I mean the one was recognizing it's an active inference podcast so starting with the kind of canonical figures that said there were some canonical figures who didn't appear on the show so that might be we could talk about that and who else could be there because there were certainly people who I think Were Meant to come on and then there were logistical timing issues and and so we can get to that but you know I think for anyone listening who wants to do something like this

people are unbelievably receptive and generous with their time um so really ultimately it was just a question of asking of course there are ways of asking so I was always you know one always has to be very polite and considerate if people say no which some people did then that's totally fine um but really it was just a question of asking so we use zencaster which is a podcasting tool that only works on Google Chrome and they've been really good I mean what's nice about them is I think they're a really small company and so whenever I had sort of little issues I would just ask them and they would get back to me super quickly so shout out to Zen caster and then I guess logistic basically the only other thing was just how big those files got so a warning I mean people who do podcasts and live streams will know I mean these were big you know 24 25 gig things and so good Wi-Fi and and an external hard drive turned out to be useful but no it's it's gone quite smoothly I'd say it's gone quite smoothly don't get me wrong it's a learning process so Daniel you'll know this there's a like one starts with no I mean right now I'm just on my laptop but people will probably be able to notice the difference right I don't have a light in front of my camera I don't have a different camera I don't have a uh microphone and so I think there were it's an optimization thing like active inference it's sort of going okay I'm starting from here and I'm accepting myself from starting here as the complete Layman the novice and I didn't really set myself a goal of where I wanted to end up but I just knew I could get better and I hope and people seem to think that the quality of it has gone up and up as time's gone on um but I think there's still massive scope and it's super cool that you guys now have charity status and hopefully that sort of raises the ceiling here and so yeah I mean personally I'm just really excited to see where this can go recognizing also the podcast face is a little is a little bit saturated and so I think when we talk about pushing this podcast forward thinking about how we can be Innovative um in setting up new schema for for guests and and and live streams throughout so yeah surprisingly quite quite a sort of fluent and and not too tricky process how how would you see it continuing like what would a very cool volunteer or intern or research fellow do or how how could they kind of jump in and continue just thematically and from the same like intention Center like what do you see that as actually being like for someone to do look I think ultimately first and foremost I would always just say speak to the people that are you're interested in and this is a sort of Trope I think in in podcasting which is you know there's a vacuum going on so I'm go hold on this is this is the issue of doing this in in in El Salvadorian host there's a lot of noise everywhere um but I think ultimately okay you know the audience is going to be it's going to be quite evidence to the audience if you're doing it for them and I'm not saying discount the interest of the

audience but I I loved everyone I spoke to so I think whoever the research uh is the podcast that is ensuring that they know who they want to speak to um is really important I think preparation was really important now I don't mean writing down all the questions you don't do this I don't mean writing down all the questions you're going to ask I always really had about three or four themes and maybe in certain cases an actual like well-formed question because it might have been slightly recursive and complicated but generally and I always said this to guests before we went live it's meant to be free flowing so I think keeping that at least for the podcast Space is really important I think that said could the community actually be even more personal you know one of the really nice things that you guys did was the with Carl which I love I love I think it's great and so I think even little stuff like that could be really nice additions to the process um because well for many people the only thing they see are their work and their Twitter account so so kind of getting to know people I think is is relevant and then yeah I mean final thing I'll say is don't I don't think we should Silo ourselves in to only speak to people that we agree with in speaking in this case um and so constantly you know whether that's people who have actually been actively critical of active inference in writing in the past so I don't know Ezekiel dealo or Evan Thompson or some like this or even someone who you know might not just be a even someone within the community right like myself who is not dispensing with the map territory distinction just having that that dialect that dialectic is really really important so yeah I think there's plenty there I mean I'd love I'd love to see it continue um I would watch every episode like an avid supporter um and yeah again again I'm just super yeah I'd be very happy to help out wherever I can but for me personally and we spoke about this it's kind of time to lock in I think um I guess it's one of those things I mean I don't know I'll ask you I think at certain point to be totally candid I was slightly worried that I was becoming a podcaster and I guess for me what I want from my academic life at least is to be writing and reading and doing academic work so I I'm curious about your experience how much do you feel like the stuff that you do takes away time for writing reading and doing the kind of canonical academic activities that's a great question so it's um there's a lot to say on it and and uh of course what I can say about my own last like 10 years doesn't apply necessarily broadly but during graduate school from 2014 to 2019 I was doing multiple weeks plus of fieldwork in the desert and while I was on Stanford campus I was not taking video meetings I was not like in the digital so during graduate school absolutely it makes sense to be really focused and then one thing I started exploring at the end of graduate school was live streaming some over head view hand drawing so with no audio so that got me familiar with OBS The

open- Source broadcasting software and just the logistics of doing something that is largely improvisational so not preparing more than oh maybe here's a reference image okay yeah so getting the time of live streaming as close to real time which is this kind of like Marshall McLuhan angle on the whole thing plus like people's feedback so that's like one sort of Lane which even now doesn't take much more time than like you mentioned several emails and and the time itself and then where it starts to really be like okay this is like reading the paper making the slides that the question is how slash is that aligned with like relevant learning and research formats so like the 0 Style video um that kind of came into play with just the pure background and context before even if we had a broader discussion on the paper that was like almost exactly the lab meeting slides which then during the second half of the lab meeting we okay go back to figure three or like what where was the abstracts like being able to have sections up there on the slide form and have one person who could say like that they had really fully gone through the whole thing so again that's the conversation with literature but from a combination of alignment of Education research service interest and then the logistical element of most streams not taking more but then there is still that element of just like what has happened um but you know that's Dan but Daniel you're you're you're a beast I mean like it's worth saying you know I think everyone's very grateful for the work you've done I mean it's like there's nothing else like it really so I mean not to be not to speak out of turn but like well done like it's really really something and I think also like to say that this is the place to go to learn about active influence I I I came to this you know I didn't this The Institute is known and I mean the the Discord is getting bigger and bigger and so yeah it's really remarkable um you know for me to be in London with not even necessarily overt active inference people and them going oh well I know there's a textbook group I know there's this chat I know that you know Daniel is doing this and that is a real sign of success um and so long may it continue I guess but yeah I I think something you said there was I forgot to mention was all the paper reading so I I hope this is quite obvious obviously one has to know what who they're talking to and so I think this in many ways informs my thing about speak to people you want to speak to now I'm quite lucky because I basically love reading academic papers but me it's just a hobby right like so it doesn't feel like work I love reading books I love read academic papers I was also really lucky and I think this is just a logistical point that I had institutional access throughout and this is a broader educational point which is that a lot of people don't have access to papers if they're not open access so I think that is probably the majority of the time goes into reading papers reading the papers of the people you're going to be speaking to and

making sure that the questions that you ask are really tied to their work because there's no point in me speaking to Ian mcgilchrist about quantum mechanics really um and so I mean I'm sure he'd have something actually quite miraculous to say but you know you get my point so yeah I you know maybe this is something that the Institute could also work on finding a way for people to access papers because for the vast majority of people who don't have a edu they can only see the abstract um but yeah on the whole yeah it's been it's been a really really Splendid experience I I mean there are I think there are plenty of people as well that we could get on um who would have lots to say awesome yeah I think maybe we could do something like see who's interested you could ask a few people and the time will be right when the time is right so there's no right need to um to jump on it I think to to ferment and digest with like the 50 hours or whatever it is there's a lot there's a lot there's a lot there that people I hope will be able to glean maybe not even insights from but just you know I think the role of philosophy in many ways is is finding the right questions to try and answer so it's not necessarily always about answering the questions it's about structuring the conversation and so hopefully we've opened up that landscape where the questions now are more concretized in people's heads the relevant questions so yeah yeah I mean I don't know hopefully an interns watching I love this internship thing that you guys are doing as well um and they're interested in taking it on because it's all theirs but again as I say whoever's watching and might be interested I am will be it although on the other side of the world right now in not the most Wi-Fi robust Place although I have found some good Wi-Fi um I'm always very happy to help always very very happy to help cool we will uh see and we'll explore it so thank you Darius for the very unique series thanks Daniel and and I also again thank you for all your help and blur and Ali and everyone who's part of the institute there's like loads and loads and loads of work that goes in that people don't see and obviously like they see my face um but yeah you guys have been incredibly helpful and generous of your time and so I'm really genuinely super super grateful thank you I think there's a piece there about like the future of content slash attention and whether and where we would really want a person like where I would want to know what you thought and where we'll be able to use synthetic method methods to just like generate different kinds of material that would flood and potentially be in distinguishable or even preferable in certain ways than than the real conversations real dialogos that people would have an experience so I think that like it you know the way that you kind of also help frame it in that context is um is awesome so thank you again well I don't although just just final thing I'll say on that I think there's something very special about two speaking to one another um and I

think I all I'll say is without the appropriate embodiment the appropriate social context the approach enculturation and so on and so forth to have genuinely the effect to genuinely not just mimic but replicate the effect that can happen when true dialogos is happening within two you know among two artificial bodies in this case I will I will I will just say that I'm skeptical but in the meantime let the humans talk and I'm sorry for sweating it's 35 degrees here so invariably there will there is this is this is not glamorous there is sweat ah and more so thank you Darius see you thanks Daniel see peace bye